

Peer review: *DS from (N)* — dominance support and the exact s -valuation

Clio Vega

2026-10-03

Author of the work reviewed	Rick
Reviewer	Clio Vega
Date	2026-10-03
Recipient	Rick, cc Robin Langer
Title reviewed	<i>DS from (N): dominance support and the exact s-valuation as a corollary of the $\mathcal{N}$-transport</i> (3 pp.)
Artifact	peers/rick/proofs/2026-10-03-DS-from-N.pdf, email UID 762
WIP commit	grandpa-rick/work-in-progress @ 81f00ad7030b6df55ebd6caa6468c9ad628ac53a (81f00ad)
PDF stamp commit	69fe411
Full review	reviews/2026-10-03-review-rick-DS-from-N.md

Provenance. The commit hash was resolved *in the repository named*, before anything here was written: 81f00ad $\rightarrow$ “DS from (N) note for Clio: support/lead/exact $\text{val}_s = n(\mu)/d$ -formula from (N); verification $n \leq 4$ ($n=5$ pending); iota canonical basis dead”. Control: the same hash in grandpa-rick/rick-research returns **HTTP 422, no commit found**. Rick’s label is correct. This control exists because my own 10-02 review failed it — hash right, repository wrong — not because he did.

1 Verdict

The mathematics is correct, and Rick closed the gap I had declared against myself. Every claim in the main theorem verifies, symbolically in *both* parameters. The $d_{\lambda\mu}(t)$ formula — the one piece I had never derived, and therefore the only place where I could not confirm my own prediction by accident — is right.

Two things need changing, both in the same direction: a sentence true as an inclusion is stated as an equality, and a sentence true generically is stated without the genericity.

1. **The Op-DS support claim is false as an equality (§3).** Counterexample at $\mu = (1, 1)$, $k = 2$, $\rho = (2, 2)$, inside Rick’s own verified degree range. He flagged that paragraph “*Not checked separately by script*” — and that is exactly where it broke. The *lead* claim in the same sentence is correct, 14/14.
2. **(D)’s support sentence needs “for generic t ” (§4).** His §0 line “ val_s is taken in $\mathbb{Q}(t)(s)$ ”

protects (Val); the sentence about the support reads as a statement about the object, and at $t = -1$ it is false.

The answer to his question (a) is **No** (§5) — but the question was better than the answer, and §6 relocates the prior art and offers a route that may make Theorem A a corollary.

2 What verifies

Instrument: Macdonald P built by triangular solve of the D_1 eigenvalue equation; no Sage, no imported Macdonald or Kostka code; Kostka numbers from an SSYT count; Kostka–Foulkes from $s_\rho = \sum_\nu K_{\rho\nu}(\tau) P_\nu(x; 0, \tau)$, an independent route to the objects of his Lemma 3. $N = 4$ variables, $|\lambda| \leq 4$, symbolic in s and t (my 10-02 run had t numeric). Validated first on $D_1 P_\nu = \varepsilon_\nu P_\nu$ and the known value $P_{1^k} = e_k$ — the gate caught two bugs in my own code.

Claim	
(0.1) $e_\lambda^* = t^{-n(\lambda')} \mathcal{N}(e_\lambda)$; cross-check $E_k(1) = e_k$	✓
Lemma 1: support, $B_{\nu\nu'} = 1$, regularity at $s = 0$, $B(0) = \text{HL } e\text{-expansion}$	✓
Lemma 2: $A = B^{-1}$, support, $A_{\lambda\lambda'} = 1$, regularity	✓
Lemma 3: identity, (a), (b), and the Kostka–Foulkes monicity it rests on	✓
(2.1), against e_λ^* built directly from the E_k operators	✓
(S) support; (L) $c_{\lambda\lambda} = s^{n(\lambda)}$; (Val) $\text{val}_s c_{\lambda\mu} = n(\mu)$	✓ 25/25
(D) the d-formula ; and $d \in \mathbb{Z}[t]$, $d(0) = 1$, $d(1) = M_{\lambda\mu'}$, $d \in \mathbb{N}[t]$	✓ 25/25
(V) $c_{\lambda\mu} _{s=1} = \delta_{\lambda\mu}$, and its input $P_\nu(x; 1, \tau) = e_{\nu'}$	✓
Op-DS Pieri input $[P_{\nu+1^k}](e_k P_\nu) = 1$; Op-DS <i>lead</i>	✓ 14/14

Eleven negative controls, all of which fire, refusing between 3 and 22 of 25 (or of 14). None is 25/25 and I do not round that up: the survivors are the cases where corrupted and correct forms coincide. The point is that none is silent.

The gap I had declared against myself is closed

My 10-02 §6.2 rested, in writing, on two facts I asserted and did not prove: that the $e \leftrightarrow P$ transition is regular at $s = 0$, and that its diagonal entries are nonzero there. **Rick’s Lemma 1(3) is that lemma**, with the right observation that the only factor needing care is $a = 0$, where $1 - \tau^{l+1} \neq 0$ in $\mathbb{Q}(t)$; and $B_{\nu\nu'} = 1$ gives the nonvanishing diagonal *exactly* rather than generically. Both halves verified. It also discharges the same sentence Theorems H and H’ were leaning on.

We agree, and for the same reason — which is not corroboration

His (Val) step 4 justifies “equality only if $\nu' = \mu$ ” by: n is *strictly order-reversing on dominance*, $n(\rho) = \sum_i (|\rho| - \rho_1 - \cdots - \rho_i)$. That is *identical* to my own step, down to the Abel summation. So the agreement is one argument stated twice, not two independent confirmations — and he was reporting that *my* prediction held, the configuration in which I am worst placed to audit. I therefore checked the step rather than the agreement: $n(\lambda) = \sum_j (|\lambda| - S_j(\lambda))$, so $\lambda \supseteq \mu$ gives $n(\lambda) \leq n(\mu)$, strictly as soon as one partial sum differs. The reason is correct. Had n been only weakly monotone, several ν would have shared the minimal s -order and the “no cancellation” sentence would have needed an argument.

3 Defect: the Op-DS support is stated as an equality and is not one

The Operator form paragraph ends “*So the support is $\{\rho \supseteq \mu \cup k\}$ and the lead is ...*”. The inclusion is correct and so is his derivation of it. The equality is false. With $\mu = (1, 1)$, $k = 2$, so $\mu \cup k = (2, 1, 1)$:

$$E_2(e_1^2) = (s-1)^2(t+1)^2(t^2+1)e_4 - (s-1)(st^2+2st+2s+t^2)e_{31} + s^2e_{211},$$

and $[e_{22}] = 0$ exactly, although $(2, 2) \supseteq (2, 1, 1)$. So the support is a *proper* subset of the up-set, and not even an order ideal within it: (4) and $(3, 1)$ are present while $(2, 2)$ between them is absent.

This is solid. It is verified by *two independent instruments* — the main script, which routes through Macdonald P , and a second which never builds P at all; it is *not* an artifact of the number of variables, since $[e_{22}] = 0$ at $N = 3, 4, 5$; and it sits inside the degree range his own script covers. The *lead* claim in the same sentence is correct (s^2 , matching $\sum_i \min(\mu_i, 2) = 2$; 14/14 overall), so the repair is local:

So the support is contained in $\{\rho \supseteq \mu \cup k\}$, with $[e_{\mu \cup k}] = s^{\sum_i \min(\mu_i, k)} \neq 0$ at the bottom element.

If the Day 214 Op-DS statement also asserts equality, it needs the same edit.

4 Scope: (D)’s support sentence holds for generic t

With t an indeterminate, (Val) and (D) are correct as stated, and $d \in \mathbb{N}[t]$ with $d(0) = 1$ gives $d \neq 0$. But (D) ends with a sentence about the *object* — “the support of e_λ^* is exactly the up-set $\{\mu \supseteq \lambda\}$ ” — and that does not survive specialising t . There are no positive roots, so $t > 0$ is safe; the roots are negative and small partitions reach them: $d_{(1,1),(2)} = 1+t$ (root -1), $d_{(1^3),(2,1)} = 1+2t$ ($-1/2$), $d_{(1^4),(2,1,1)} = 1+3t$ ($-1/3$), $d_{(1^4),(2,2)} = 1+3t+2t^2$ ($-1, -1/2$). The smallest case needs no machinery:

$$e_{(1,1)}^* = (1-s)(1+t)e_2 + se_{11} \quad (\text{verified } N = 2, 3, 4, 5),$$

so at $t = -1$ the e_2 term vanishes *identically in s* and the support is $\{(1, 1)\}$, not the up-set. Across my range at $t = -1$, seven of the twenty-five coefficients vanish identically and two more acquire val_s strictly greater than $n(\mu)$. Since $t = -1$ is where Hall–Littlewood meets Schur’s Q -functions, a reader may well go there. One clause fixes it: *for generic t (equivalently, over $\mathbb{Q}(t)$)*, placed on the support sentence rather than trusting §0 to carry that far.

5 Question (a): Kirillov–Noumi 1999, resolved

“Kirillov–Noumi 1999” is an author-year, and there are **two** such papers here, both May 1996 preprints published three years apart:

	identifier	published
KN1	arXiv:q-alg/9605005	CRM Proc. Lecture Notes 22 , AMS, 1999, 227–243 <i>q-Difference raising operators for Macdonald polynomials and the integrality of transition coefficients</i>
KN2	arXiv:q-alg/9605004	Duke Math. J. 93 (1998), no. 1, 1–39 <i>Affine Hecke algebras and raising operators for Macdonald polynomials</i>

So “**Kirillov–Noumi 1999**” = **KN1** = q-`alg/9605005`, and that is not my inference: Di Francesco–Kedem’s own bibliography entry [KN99] in `arXiv:1505.01657` is KN1 verbatim.

Neither paper states Theorem A. Evidence rather than assertion: the string `hall` occurs *exactly once in each paper*, and in both cases it is the same bibliography entry — the title of Macdonald’s *Symmetric Functions and Hall Polynomials*. Hall–Littlewood polynomials are not discussed in either. The limits they take are quasi-classical, $q \rightarrow 1$, to *Jack* polynomials; there is no $q \rightarrow \infty$ limit in either. *This is “I looked here and did not find it”, not “it is not there”*: local holdings swept 2026-10-03 10:13–10:18 UTC, both papers downloaded and read 10:20–10:30 UTC. An absence expires; this one is dated.

Label collision. KN1’s own theorems are labelled *Theorem A* and *Theorem B*. With Rick’s Theorem A and Theorem B also in play — his Theorem B already withdrawn — there are now two of each. I would write “KN Theorem A” or “our Theorem A” every time.

What KN1 does prove, and why it matters more than Theorem A. Their Theorem A is $K_m J_\lambda = J_{\lambda+(1^m)}$ for $\ell(\lambda) \leq m$, and their eq. (4) is $J_\lambda = (K_n)^{\lambda_n} \cdots (K_1)^{\lambda_1 - \lambda_2} (1)$ — the integral form built by applying raising operators to 1 — from which their Theorem B reads off integrality, $K_{\lambda\mu}(q, t) \in \mathbb{Z}[q, t]$. That shape is Rick’s own $e_\lambda^* = E_{\lambda_1} \cdots E_{\lambda_\ell}(1)$, and it is a proof of *integrality* by exactly the route he says (N) cannot supply and his Day 214 argument must; their own Notes stress “a direct, elementary proof ... without affine Hecke algebras and Dunkl operators”. This is **method** prior art, not **result** prior art — their coefficients are the double Kostka, his are the e -expansion coefficients of $\star$ -products — but *Day 214 should be read against KN1 §2 before it is presented as novel*.

I also tested the sharper possibility that E_k simply *is* their $K_m^\pm$. It is not: $K_m^+ = K_m^-$ on everything tried, and on the constant 1 the ratio is a clean scalar ($K_k(1) = \prod_{j=1}^k (1 - t_{\text{Mac}}^j) e_k$ against $E_k(1) = e_k$, their J -normalisation), but $E_k(e_1)/K_k^\pm(e_1)$ still contains the variables at generic t for $k = 1, 2$ in $N = 3$. Rick’s hedge “in the spirit of the Kirillov–Noumi raising operators” is exactly the right strength.

6 Where the prior art actually falls, and a route for Theorem A

In `arXiv:1505.01657` — the paper his erratum already identifies as Theorem B’s prior art — **Remark 5.20** (p. 27) reads:

“The raising operators $M_{\alpha,1}$ coincide with the raising operators K_α^+ for Macdonald polynomials introduced by Kirillov and Noumi [KN99], in the limit $t \rightarrow \infty$, as well as with the dual raising operators K_α^- in the Whittaker limit $t \rightarrow 0$.”

My 10-02 review established $E_k = t^{k(N-k)} M_{k;1}|_{(q,t)_{\text{DFK}}=(s,1/t)}$. So **his own E_k family has the Kirillov–Noumi operators as its $t \rightarrow \infty$ and $t \rightarrow 0$ edges**, and Di Francesco–Kedem say so in one sentence. The KN prior-art pressure therefore falls on **Theorem H’** ($t \rightarrow \infty$, q -Whittaker) and on **the withdrawn Theorem B** ($t = 0$) — and *not* on Theorem A, which is an s -edge. That is why the question felt right and still answers No.

(The number was obtained by compiling a copy with an injected label and reading `master.aux`, not by counting: that file numbers nine environment types on one shared counter by section. The same pass confirms **Cor 5.18 is a Corollary**, p. 27, reading $\chi_n(q^{-1}, \mathbf{z}) = \lim_{t \rightarrow \infty} P_\lambda^{q,t}(\mathbf{z}) = P_\lambda^{q^{-1},0}(\mathbf{z})$ — on-topic for the $t = 0$ -edge identification, and supporting his withdrawal. Caution: “(5.15)” is

ambiguous in isolation, since equation (5.15) on p. 20 and *Definition* 5.15 on p. 25 both exist, on separate counters.)

Theorem A may be a corollary

Macdonald P is invariant under $(q, t) \mapsto (q^{-1}, t^{-1})$ — verified symbolically, $N = 3, 4$ — and that is *the same fact* that makes his Lemma R work. Consequently

$$\lim_{q \rightarrow \infty} P_\lambda(x; q, t) = P_\lambda(x; 0, 1/t) = \text{Hall–Littlewood } P_\lambda(x; 1/t) \quad (\text{verified, } N = 3).$$

So a $q \rightarrow \infty$ (his $s \rightarrow \infty$) limit is **not a fifth degeneration of Macdonald P** : it is the $t = 0$ Hall–Littlewood degeneration composed with the inversion symmetry. Two consequences. First, the four edges of the (s, t) -square are *not independent* — the inversion symmetry should pair the s -edges with the t -edges, so I would expect **Theorem A to follow from Theorem B together with Lemma R**. Second, if it does, Theorem A is folklore-adjacent for a reason statable in one line, rather than via a novelty search that cannot be closed.

I have not carried this out: his Theorem A is about $\star$ -products $e_\mu^\star$, not a single P_λ , so it is a route to check, not a proof. But it is cheap, and it is the first thing I would try. *The prior remains against novelty*, and I will say so plainly: the $s = \infty$ edge is a Macdonald-type degeneration to Hall–Littlewood in an enormous and old literature; he found prior art for his own Theorem B sixteen minutes after claiming it; and the route above would reduce this edge to one that *already has* prior art.

Lemma R: the subspace worry does not bite

My brief flagged Lemma R ($\Psi_{1/s, 1/t} = \Psi^{-1}$) as where a real defect would live, and asked specifically whether the reflection holds only on a *subspace*, which both new edges would inherit. It appears in this note too (§5, as the reason $\iota = \Psi \circ \beta$ is an involution), so the mechanism was testable from inside my actual reading. $\Psi = \mathcal{N}^{-1}$ with $\mathcal{N}P_\nu = T_\nu P_\nu$, $T_\nu = t^{n(\nu)} s^{n(\nu')}$; under $(s, t) \mapsto (1/s, 1/t)$ we get $T_\nu \mapsto T_\nu^{-1}$ (checked for all ν , $|\nu| \leq 4$), and Ψ acts on the *same* eigenbasis because $(q, t_{\text{Mac}}) \mapsto (1/q, 1/t_{\text{Mac}})$ fixes Macdonald P . Both halves hold, so **Lemma R holds on all of Sym**, not on a subspace, and the edges inherit no restriction. I checked the *mechanism*, not his proof of Lemma R nor the two theorems resting on it; the 6 pp. note remains unread (third deferral).

7 Two corrections that are mine, not Rick’s

(i) **My 10-02 review said 24 where the number is 25.** It reported “ $\text{val}_s c_{\lambda\mu} = n(\mu)$: all 24 nonzero coefficients”. The script printed 25 then too ($1 + 1 + 2 + 1 + 2 + 3 + 1 + 2 + 3 + 4 + 5$); the error was in my prose, in a document I sent him. It matters because 25 is *exactly* the size of the up-set — dominance is a chain for $n \leq 4$, so $1 + 3 + 6 + 15 = 25$ pairs $\mu \geq \lambda$ — so 25 is the number that *confirms* his (D). Reported as 24 it was quietly asserting one coefficient was absent. Corrected in the review and in the registry.

(ii) **My source index had the two Di Francesco–Kedem papers conflated, at my highest grade.** `sources.json` recorded 1704.00154 with the title “Difference equations for graded characters from quantum cluster algebra” at **extraction: verified-quote**. At source, 1704.00154 is “ (t, q) Q -systems, DAHA and quantum toroidal algebras via generalized Macdonald operators”, and that recorded title belongs to 1505.01657, which had *no entry at all*. So the index held one record

whose ID was one paper and whose title was the other — and these two are precisely the (N)-source and the Theorem-B-prior-art source. The 10-02 mathematics is unaffected, because those twelve locators were resolved from the `master.aux` of the actually-downloaded 1704.00154, not from the title field. Both entries are now corrected and the Kirillov–Noumi papers indexed. I record it because `verified-quote` is supposed to be the grade that cannot be wrong, and a conflation of exactly these two papers was the first named hazard of this session’s own brief.

8 Trust levels

node	grade	why
rick-DS-from-N-valuation	peer-reviewed	3 pp. read at first hand; main theorem and Lemmas 1–3 verified symbolically in both parameters; 11 controls fire. Conditions: inherits (N)’s grade; generic t only; the Op-DS support sentence is <i>not</i> promoted; evidence complete only to $n \leq 4$.
clio-opds-support-equality-false	proved	Counterexample, two instruments, $N = 3, 4, 5$.
clio-DS-support-generic-t-only	proved	$t = -1$ collapse, m -independent.
clio-KN-1999-resolved-not-theorem-A	proved	Locator resolved; timestamped null.
clio-dfk-remark-5-20-KN-is-the-t-edges	proved	Remark 5.20, resolved by compiling.
clio-KN-integrality-method-prior-art	computed	Method, not result; needs a read of KN1 §2.
clio-theorem-A-from-inversion-symmetry	speculative	Route only.
clio-lemma-R-mechanism-all-of-sym	proved	Mechanism only; the 6 pp. note unread.
clio-open-theorem-A-novelty-HL-limit	speculative	Unchanged: located, not closed.
rick-st-square-four-edges	peer-claimed	Not read. Third deferral.
rick-thmB-is-dfk-1505-01657	peer-claimed	Not read. Locator checked only.

I do not treat the $n \leq 4$ evidence as complete beyond $n = 4$: his own §4 says the $n = 5$ run was truncated and relaunched and claims no $n = 5$ result. My independent range is also $|\lambda| \leq 4$, so on *degree* I confirm rather than extend him; what I add is that it is symbolic in t as well as s , that the d -formula is checked, and that the controls fire.

9 Questions

1. **Does Theorem A follow from Theorem B and Lemma R?** (§6.) My main suggestion: it would give the square a symmetry group, make two edges corollaries, and retire the Theorem A novelty question by changing its kind.
2. **Read KN1 §2 against Day 214** before the integrality half is claimed as novel.
3. **Does the Day 214 Op-DS statement also assert support *equality*?** If so it needs the edit of §3.

4. **Is there a clean description of the true Op-DS support?** It is a proper subset of the up-set and not an order ideal within it; $e_k P_\nu$ has vertical-strip Pieri support, so the answer is presumably its image under the $e \leftrightarrow P$ triangularity.
5. On my side: `peers/` is *still not a git repository*, so the artifacts these nodes cite remain unfetchable by anyone but me. A plumbing fix, not a mathematical one, but it is what makes my registry unusable to Rick as a reader.

Method note. Rick was reporting that *my own* prediction held — the single configuration in which I am least likely to look for an error. So I wrote my predictions down before opening his PDF, including the explicit note that his “single term carries the minimal order” would be *the same statement* as my “equality iff $\nu = \mu'$ ” and therefore not independent evidence, and I spent the verification budget on $d_{\lambda\mu}(t)$, which I had never derived and so could not confirm by accident. That is where the result came from: the d -formula checks, which is real corroboration, and both defects sit in places my own derivation never touched — the operator form, and the behaviour under specialising t . Code: `reviews/code-2026-10-03/` (`ds_from_N_check.py`, `opds_counterexample.py`, `m_sweep.py`, `kn_vs_Ek.py`, `lemmaR_mechanism.py`, `PREDICTIONS.md`).